. From the definition and the examples, it is clear that the 10's complement of a decimal number can be formed by leaving all least significant zeroes unchanged, subtracting the first non-zero least significant digit from 10, and then subtracting all other higher significant digits from 9.

. The 2's complement can be formed by leaving all least significant zeroes and the first nonzero digit unchanged, and then replacing 1's by 0's and 0's by 1's in all other higher significant digits.

. The r's complement of a number exists for any base r (r greater than but not equal to 1) and may be obtained from the definition given above.

1.5.2 The (r-1)'s Complement

. Given a positive number N in base r with an integer part of n digits and a fraction part of m digits, the (r-1)'s complement of N is defined as $r^n - r^{-m} - N$.

Example: The 9's complement of $(52520)_{10}$ is $(10^5 - 1 - 52520) = (99999 - 52520) = 47479$

. No fraction part, so $10^{-m} = 10^0 = 1$

. The 9's complement of $(0.3267)_{10}$ is $(1 - 10^{-4} - 0.3267) = 0.9999 - 0.3267 = 0.6732$

. No integer part, so $10^n = 10^0 = 1$

. The 9's complement of $(25.639)_{10}$ is $(10^2 - 10^{-3} - 25.639) = 99.999 - 25.639 = 74.360$.

. The 1's complement of $(101100)_2$ is $(2^6 - 1) - (101100) = (111111 - 101100)_2 = 010011$.

. The 1's complement of $(0.0110)_2$ is $(1 - 2^{-4})_{10} - (0.0110)_2 = (0.1111 - 0.0110)_2 = 0.1001$.

. The 7's complement of $(76)_8$ is $(8^2 - 8^0)_{10} - 76_8 = (63)_{10} - 76_8 = (77 - 76)_8 = 1_8$.

From the examples, we see that the 9's complement of a decimal number is formed simply by subtracting every digit from 9. The 1's complement of a binary number is even simpler to form: the 1's are changed to 0's and the 0's to 1's.

The r's complement can be obtained from the (r-1)'s complement after the addition of r^{-m} to the least significant digit.

For example, the 2's complement of 10110100 is obtained from the 1's complement 01001011 by adding 1 to give 01001100.